



Department of Computer Science
VERITAS UNIVERSITY, ABUJA
Bwari Area Council,
Website: <http://www.veritas.edu.ng>

CSC 411 Research Methods

Lecture 5: Data Analysis and Interpretation

Learning Objectives

At the end of this lecture, students should be able to:

1. Understand the concept and importance of data analysis in research.
2. Apply appropriate techniques for analyzing quantitative and qualitative data.

1. Introduction

Data analysis and interpretation are critical stages in the research process. After data collection, researchers must organize, analyze, and interpret data in order to draw meaningful conclusions and support or reject research hypotheses. This lecture focuses on both quantitative and qualitative data analysis techniques, as well as how to present and interpret findings effectively.

2. Qualitative Data Analysis

Qualitative data refers to non-numeric information such as interview transcripts, notes, video and audio recordings, images, and text documents. Qualitative data analysis can be divided into the following five categories:

1. Content analysis. This refers to the process of categorizing verbal or behavioural data to Classify, summarize, and tabulate the data.

2. Narrative analysis. This method involves the reformulation of stories presented by respondents taking into account the context of each case and the different experiences of each respondent. In other words, narrative analysis is the revision of primary qualitative data by the researcher.

3. Discourse analysis. A method of analysis of naturally occurring talk and all types of written text.

4. Framework analysis. This is a more advanced method that consists of several stages, such as familiarization, identifying a thematic framework, coding, charting, mapping, and interpretation.

5. Grounded theory.

This method of qualitative data analysis starts with an analysis of a single case to formulate a theory. Then, additional cases are examined to see if they contribute to the theory.

Qualitative data analysis can be conducted through the following three steps:

Step 1: Developing and Applying Codes. Coding can be explained as the categorization of data. A ‘code’ can be a word or a short phrase that represents a theme or an idea. All codes need to be assigned meaningful titles. A wide range of non-quantifiable elements such as events, behaviours, activities, meanings etc. can be coded.

There are three types of coding:

1. *Open coding.* The initial organization of raw data to try to make sense of it.
2. *Axial coding.* Interconnecting and linking the categories of codes.
3. *Selective coding.* Formulating the story through connecting the categories.

Coding can be done manually or using qualitative data analysis software such as

NVivo, Atlas ti 6.0, Hyper RESEARCH 2.8, Max QDA, and others.

When using manual coding, you can use folders, filing cabinets, wallets, etc., to gather together materials that are examples of similar themes or analytic ideas. The manual method of coding in qualitative data analysis is rightly considered labor-intensive, time-consuming, and outdated.

In computer-based coding, on the other hand, physical files and cabinets are replaced with computer-based directories and files. When choosing software for qualitative data analysis, you

need to consider a wide range of factors such as the type and amount of data you need to analyse, time required to master the software and cost considerations.

Moreover, it is important to get confirmation from your dissertation supervisor prior to application of any specific qualitative data analysis software.

The following table contains examples of research titles, elements to be coded and identification of relevant codes:

Qualitative data coding

Research title	Elements to be coded	Codes
Born or bred: revising The Great Man theory of leadership in the 21 st century	Leadership practice	Born leaders Made leaders Leadership effectiveness
A study into advantages and disadvantages of various entry strategies to the Namibian clothing market	Market entry strategies	Wholly-owned subsidiaries Joint-ventures Franchising Exporting Licensing
Impacts of CSR programs and initiative on brand image: a case study of Coca-Cola Company Namibia.	Activities, phenomenon	Philanthropy Supporting charitable courses Ethical behaviour Brand awareness Brand value
An investigation into the ways of customer relationship	Tactics	Viral messages

management in mobile marketing environment		Customer retention Popularity of social networking sites
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Step 2: Identifying themes, patterns, and relationships.

Unlike quantitative methods, in qualitative data analysis, there are no universally applicable techniques that can be applied to generate findings. The analytical and critical thinking skills of a researcher plays significant role in data analysis in qualitative studies. Therefore, no qualitative study can be repeated to generate the same results.

Nevertheless, there is a set of techniques that you can use to identify common themes, patterns, and relationships within responses of sample group members in relation to codes that have been specified in the previous stage.

Specifically, the most popular and effective methods of qualitative data interpretation include the following:

- **Word and phrase repetitions** – scanning primary data for words and phrases most commonly used by respondents, as well as, words and phrases used with unusual emotions;
- **Primary and secondary data comparisons** – comparing the findings of interview/focus group/observation/any other qualitative data collection method with the findings of the literature review and discussing differences between them;
- **Search for missing information** – discussions about which aspects of the issue were not mentioned by respondents, although you expected them to be mentioned;
- **Metaphors and analogies** – comparing primary research findings to phenomena from a different area and discussing similarities and differences.

Step 3: Summarizing the data. At this last stage, you need to link research findings to hypotheses or the research aim and objectives. When writing the data analysis chapter, you can use noteworthy quotations from the transcript in order to highlight major themes within findings and possible contradictions.

It is important to note that the process of qualitative data analysis described above is general and different types of qualitative studies may require slightly different methods of data analysis.

3.0 Quantitative Data Analysis

In quantitative data analysis, you are expected to turn raw numbers into meaningful data through the application of rational and critical thinking. Quantitative data analysis may include the calculation of frequencies of variables and differences between variables. A quantitative approach is usually associated with finding evidence to either support or reject hypotheses you have formulated at the earlier stages of your research process.

The same figure within a data set can be interpreted in many different ways; therefore, it is important to apply fair and careful judgment.

For example, questionnaire findings of a research titled “A study into the impacts of informal management-employee communication on the levels of employee motivation: a case study of Agro Bravo Enterprise” may indicate that the majority 52% of respondents assess communication skills of their immediate supervisors as inadequate.

This specific piece of primary data findings needs to be critically analyzed and objectively interpreted by comparing it to other findings within the framework of the same research. For example, the organizational culture of Agro Bravo Enterprise, leadership style, the levels of frequency of management-employee communications need to be taken into account during the data analysis.

Moreover, literature review findings conducted at the earlier stages of the research process need to be referred to in order to reflect the viewpoints of other authors regarding the causes of employee dissatisfaction with management communication. Also, secondary data needs to be integrated in data analysis in a logical and unbiased manner.

Let’s take another example. You are writing a dissertation exploring the impacts of foreign direct investment (FDI) on the levels of economic growth in Vietnam using a quantitative data analysis method. You have specified FDI and GDP as variables for your research and correlation tests produced correlation coefficient of 0.9.

In this case simply stating that there is a strong positive correlation between FDI and GDP would not suffice; you have to provide explanation about the manners in which the growth on the levels of FDI may contribute to the growth of GDP by referring to the findings of the literature review and applying your own critical and rational reasoning skills.

A set of analytical software can be used to assist with analysis of quantitative data. The following table illustrates the advantages and disadvantages of three popular quantitative data analysis software: Microsoft Excel, Microsoft Access and SPSS.

Advantages and disadvantages of popular quantitative analytical software.

	Advantages	Disadvantages
<i>Excel Spreadsheet</i>	Cost effective or Free of Charge Can be sent as e-mail attachments & viewed by most smartphones All in one program Excel files can be secured by a password	Big Excel files may run slowly Numbers of rows and columns are limited Advanced analysis functions are time consuming to be learned by beginners Virus vulnerability through macros
<i>Microsoft Access</i>	One of the cheapest amongst premium programs Flexible information retrieval Ease of use	Difficult in dealing with large database Low level of interactivity Remote use requires installation of the same version of Microsoft Access
<i>SPSS</i>	Broad coverage of formulas and statistical routines Data files can be imported through other programs Annually updated to increase sophistication	Expensive cost Limited license duration Confusion among the different versions due to regular update

Quantitative data analysis with the application of statistical software consists of the following stages:

1. Preparing and checking the data. Input of data into a computer.
2. Selecting the most appropriate tables and diagrams to use according to your research objectives.
3. Selecting the most appropriate statistics to describe your data.
4. Selecting the most appropriate statistics to examine relationships and trends in your data.

It is important to note that while the application of various statistical software and programs is invaluable to avoid drawing charts by hand or undertaking calculations manually, it is easy to use them incorrectly. In other words, quantitative data analysis is “a field where it is not at all difficult to carry out an analysis which is simply wrong, or inappropriate for your data or purposes. And the negative side of readily available specialist statistical software is that it becomes that much easier to generate elegantly presented rubbish”.

Therefore, it is important for you to seek advice from your dissertation supervisor regarding statistical analyses in general and the choice and application of statistical software in particular.